7.10. Historical interlude: Eilenberg, Mac Lane, and "general abstract nonsense"

Saunders Mac Lane's (1909–2005) father and grandfather were both Congregational ministers, so young Saunders seemed destined for an ecclesiastical career. Early on, however, he started having "various theological doubts and questions". "[A] ministerial career would have required my belief of things of which I was uncertain", Mac Lane later recalled. "Mathematics, however, provided a different sort of certainty". One thing he was certain about was that the minister's income was woefully inadequate to provide for comfortable retirement. As a result, Mac Lane decided to pursue a career that would be "scientific rather than ministerial" [**33**, pp. 349–350]. He earned his bachelor's degree at Yale and his master's at Chicago.

Inspired by the "elegance of German abstract algebra", in which the notions of rings and fields axiomatized addition and multiplication, Mac Lane devoted his master's thesis to a similar axiomatization of exponentials, as well as plus and times. At Chicago, Mac Lane was put off by inadequate teaching and by examples of what he saw as a "display of pedantry", and he concluded that the mathematics department there "had ceased to be really first class" [**36**, p. 152]. He recalled that he had been taught that a vector was an n-tuple, "an idea that I soon had to unlearn" [**33**, p. 39]. Mac Lane wanted to write a doctoral dissertation on logic but could not find an advisor and decided to leave for Göttingen, the preeminent center of mathematical logic research.

At Göttingen, the home of the famous Hilbert school, Mac Lane learned group representations from Hermann Weyl and linear associative algebras from Emmy Noether. "To live in Göttingen was to be immersed in mathematical culture", Mac Lane wrote, recalling Göttingen as "an amalgam of research, teaching, and inspiration". Despite this immersion, Mac Lane could hardly ignore the tense political climate, the vociferous clashes among 27 political parties, and the students dressed in the Nazi brownshirt uniform. Mac Lane did write a thesis on logic but managed to overlook Gödel's famous incompleteness theorem, which appeared at that time. His thesis elicited